SAINT MICHAEL COLLEGE OF CARAGA

THE IMPACT OF 2021 MODERNIZATION PROGRAMS AS PERCEIVED BY THE BFP PERSONNEL

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CHAPTER 1

THE PROBLEM AND ITS SCOPE

Background of the study

During emergencies, fire protection seeks to protect people, property, and the environment. The Bureau of Fire Protection (BFP) is responsible for this endeavor in the Philippines. Rescue operations, fire prevention, and incident response are among the agency's primary responsibilities. It also looks into fire incidents and enforces fire safety rules. It also works with volunteer fire brigades to strengthen coordination and improve community preparedness.

However, the BFP continues to experience several long-standing challenges. Many stations still use old equipment, lack enough personnel, and operate with limited resources. The agency's capacity to respond promptly and provide uniform fire protection nationwide is frequently hampered by these issues. The government implemented modernization measures to improve the BFP's overall performance and operations and address these problems. The BFP Modernization Act of 2021 focuses on three main goals. First, it upgrades fire stations and other facilities. Second, it improves emergency response systems. Third, it strengthens coordination between BFP staff and volunteer firefighters. Evaluating the program is important to see if it works effectively and can be maintained over time (MJBAS, 2022).

International experience demonstrates that modernization efforts in fire services significantly improve emergency response performance. In the United States, Japan, and Australia, studies show that holding regular training sessions and improving communication systems helps lower fire-related deaths and limit property damage (Smith & Jones, 2020; Kim et al., 2021).

In the United Kingdom, firefighters now use modern equipment along with digital command systems. This combination improved station coordination. Fire stations responded to emergencies faster and coordinated more effectively as a result (Johnson, 2019).

Similarly, in Singapore, modernization efforts have emphasized interoperability between government fire services and volunteer brigades, resulting in a more unified and efficient response system. These international experiences provide valuable insights into the potential outcomes of modernization initiatives in developing nations.

In the Philippines, fire service modernization became more critical after Republic Act No. 11589, or the BFP Modernization Act of 2021. This law calls for building new fire stations, upgrading facilities, and buying modern fire trucks and equipment. Local studies indicate these programs have improved operations in some areas, but challenges persist. Training, maintenance, and the equitable distribution of resources between urban and rural stations still require improvement. Volunteer fire brigades, who serve as essential partners, also report mixed levels of support. While some have access to shared resources, others still lack adequate assistance (Dela Cruz, 2022; Reyes, 2023).

Although the modernization program aims to make the BFP a world-class fire service, little research has examined how staff and volunteers feel about the changes. This study addresses that gap. It examines whether modernization meets operational needs, improves skills, and strengthens cooperation between government and volunteer teams. It looks at equipment, training, organizational efficiency, and morale. By identifying both strengths and weaknesses, the study provides practical guidance for better planning and resource management.

This research is helpful for many groups. Policymakers and government agencies can assess the effectiveness of modernization investments and identify areas for improvement. The BFP can use the results to improve training, maintain equipment, and optimize operations. Finally,

the study adds to public safety research and can guide future fire service programs. Overall, the goal is to strengthen fire protection, make communities safer, and increase public trust in emergency services.

Review of Related Literature

Before this study, the chapter reviews existing literature. It looks at articles, books, and dissertations. The goal is to understand the topic and guide the research.

Relate Literature

The BFP Modernization Act of 2021 (Republic Act No. 11589) aimed to strengthen the Bureau of Fire Protection. It upgraded facilities, staff, and equipment. It also established specialized services, including emergency medical response and HAZMAT operations. The BFP modernization program includes fire protection services, force restructuring, organizational development, capacity building, technology upgrades, specialized services, and human resources development (Lawphil, 2021).

Santos et al. (2024) reported improvements in equipment, technology, training, and facilities nationwide. Modern equipment helps staff respond faster. Training improves skills and job satisfaction. Upgraded facilities and communication systems boost morale, safety, and community involvement (Department of Budget and Management, 2023; Cruz et al., 2023).

These areas are key parts of the modernization program. Legislative documents and recent studies confirm their importance.

The ten-year program aims to make the BFP more effective and keep communities safer. Scholars say modernization is not only about equipment. It also includes training, morale, and efficiency (Delarosa, 2021). Organizational reforms increase staff competence and motivation when they fit field needs (Alonzo, 2019).

The program also values fire volunteers. Volunteers help the BFP reach more communities. Training them and including them in local programs improves disaster preparedness and public trust (Santos, 2020).

Studies in other countries show the same. Volunteers included in training feel more involved and contribute more to safety (Martinez, 2018). These show that both staff and volunteers are important for success. Some studies warn that modernization may face challenges. Bureaucracy, uneven resources, and volunteers paying for fuel and equipment reduce effectiveness (Reyes, 2022).

Age

The age profile of BFP personnel helps us understand how the 2021 Modernization Program affects them. Most BFP personnel in the Philippines are young, mostly aged 21 to 40 (Bautista & Rodriguez, 2022).

In Laguna province, almost one-third (29.79%) of personnel aged 21 to 30 made up one group. Another large group included those aged 31 to 35. These indicate that all staff members are in the early to mid-career stage. Younger personnel adapt more easily to new technologies and updated procedures (EPRA International Journal of Multidisciplinary Research, 2023).

In Sorsogon province, most personnel fell within the 25-34 age range. In the National Capital Region, many were 30 to 34. This mix of individual insight and experience helps balance adaptability with knowledge. Most staff in these age groups are male, reflecting traditional roles in firefighting. Female participation is growing. The age profile shows why training and capacity-building programs should focus on staff who can benefit most from modernization (All Multidisciplinary Journal, 2023).

Heraldero & Lomarda (2021) studied BFP personnel in Bukidnon. They found that staff aged 21 to 35 were more engaged in training programs, like the Fire Basic Recruit Course and leadership seminars. Younger personnel were mainly male and new to the service. They should

be ready to use new procedures and technologies. Continuous training improves fire prevention and suppression performance.

Looking at these age groups helps assess the program's effectiveness and sustainability. A younger workforce can adopt innovations, follow new procedures, and maintain the gains from modernization. This profile fits the national goal of improving public safety through technology and training. Age remains an important factor for planning future programs in the BFP (Scribd Firefighter Profile, 2025).

Gender

Understanding the gender profile of respondents, especially BFP personnel and volunteer firefighters, is important for understanding how the 2021 Modernization Program affects people. Studies consistently show that the workforce is primarily male, but female participation is increasing (Civil Service Commission, 2022).

A study in Sorsogon Province found that about 70% of BFP personnel were male, while females accounted for about 30% of the workforce. These show that firefighting has been male-dominated, but agencies are making efforts to include more women. Female personnel hold various ranks and contribute significantly to firefighting, emergency medical services, and disaster management (Multidisciplinary International Journal of Research and Development, 2024).

The modernization program highlights gender-specific roles and supports female personnel. For example, specialized training and leadership programs help the careers of female fire officers take on greater responsibilities and leadership positions. These initiatives aim to remove gender barriers and give women equal opportunities for career growth. This approach aligns with national goals for gender equality and the integration of women into traditionally male

fields such as firefighting and emergency response (Multidisciplinary International Journal of Research and Development, 2024).

A study in Dagupan City shows that male personnel still dominate operational roles, and female participation is limited. Gender balance varies across locations, showing the need for local strategies to include more women. Recruitment, training, and awareness programs can help ensure FEMA personnel are more fairly represented and involved in modernization efforts across the board (EPRA International Journal of Multidisciplinary Research, 2023).

In Cavite, research shows that although firefighting is still predominantly male, the number of female firefighters has increased, especially in administrative, training, and support roles. This change reflects a shift in organizational norms encouraged by the modernization program, which promotes gender equality. Inclusion policies and gender sensitivity training are important parts of this transformation to modernize the BFP's workforce culture (Scribd, 2025).

International studies also show the benefits of gender inclusivity in fire services. In the United Kingdom, including women in fire and rescue services led to more teamwork and better communication (Anderson, 2020). In South Korea, female firefighters play strong leadership roles in community programs and fire prevention education (Kim, 2021). These studies suggest that gender balance not only supports equality but also improves efficiency and service delivery, which the BFP is gradually adopting as part of its modernization agenda.

In the Philippines, several studies show how gender dynamics in the BFP are changing. Reyes and Dela Cruz (2022) found that the number of women in technical and administrative roles increased after modernization introduced gender-sensitive training and fair promotion systems. Morales (2021) noted that gender inclusivity in recruitment improves public trust, mainly when female firefighters serve as approachable role models. These findings underscore the role of

gender equality in enhancing professionalism and fostering greater public engagement within the BFP.

Rank

According to Sanidad-Leones et al. (2023), their study examined perceptions of the impacts of BFP modernization in the Caraga Region and found that rank and role influenced views on improvements in fire protection services, organizational development, and training effectiveness. Higher ranks perceived greater strategic benefits, while frontline personnel emphasized operational improvements. They also highlighted the importance of continuous training and equipment upgrades.

Organizational-change research suggests that perceptions of such reforms often vary by rank or hierarchical level. For example, Jones et al. (2018) reported that executives maintained a generally positive outlook during a hospital restructuring. In contrast, supervisory and rank-and-file staff referred more to conflict and divisions and expressed more negative attitudes toward the change than did executives.

Likewise, Wynen et. al (2022) report that employees at higher (strategic) levels tend to view organizational reforms more favorably than lower-level or technical staff. These studies indicate that higher-ranking officers often appreciate the goals of modernization more than those closer to frontline duties.

By contrast, mid-level supervisors and ordinary personnel are more likely to focus on the challenges or uncertainties introduced by large-scale changes. Sunada et al. (2025) evaluated the competencies of BFP personnel in the National Capital Region and found that modernization has improved the capabilities of fire officers across ranks. They emphasized that training programs

enhanced personnel skills in firefighting, rescue, and fire prevention, leading to improved performance aligned with rank responsibilities.

Moreover, Kim et al. (2025) examined Korean firefighters' views during a conversion to the national civil service. They found that overall satisfaction with the reform increased with rank within the fire organization. Lower-ranking roles (such as paramedics in their study) reported lower satisfaction compared to line firefighters. In the context of modernization, rank-and-file FP personnel, or those in lower-tier roles, may be less optimistic than senior officers. These findings underscore that rank and position can shape how employees experience and evaluate organizational change: upper echelons often have a broader perspective on objectives, while lower ranks may be more sensitive to operational impacts.

Differences between career personnel and volunteers are also important. Studies of fire services note that volunteers can react quite differently to reforms than paid staff. Van den Ende et al. (2020) studied Dutch fire service regionalization and found "notable tensions" between professional managers and volunteer firefighters during the reform. Their analysis highlights a clash between the drive for professionalization and volunteers' local, intrinsic motivations. In practical terms, volunteer BFP firefighters (who are numerous in the Philippines) may have distinct concerns or sense-making about modernization, for instance, questioning whether new systems adequately support their role or fearing loss of traditional autonomy.

Furthermore, Ramos et al. (2025) highlighted that personnel in higher ranks possess advanced decision-making and leadership skills, enabling them to implement modernization initiatives effectively. In contrast, lower-ranking personnel demonstrate strong operational skills but may require additional training to align with modernization objectives fully.

Similarly, Lomarda (2025) found that age and experience—demographic factors that often correlate with rank—influence the performance and perceptions of BFP personnel, with younger personnel demonstrating greater effectiveness in fire suppression and prevention tasks. Longa (2024) emphasized that tailored training programs should consider rank and length of service, as personnel at different ranks exhibit varying capabilities and experiences that can impact the success of modernization efforts.

Moreover, Sabalosa (2024) noted that higher-ranking personnel, such as station chiefs, face unique challenges and responsibilities that provide them with distinct insights into the program's implementation, reinforcing the importance of rank in evaluating the modernization program's overall effectiveness. Collectively, these studies suggest that rank is a critical factor in understanding how BFP personnel and fire volunteers perceive the modernization program, and addressing rank-specific needs can enhance the program's impact and sustainability.

Years of service

The study by Lomarda (2025) shows that years of service affect how BFP personnel see modernization. Lomarda studied personnel in the Southern Districts of Bukidnon. The study found that younger personnel performed well in fire suppression and prevention. It also found a link between fire skills and overall performance. Longer-serving personnel use their experience to shape their views on reforms. Their practical knowledge and loyalty influence how they respond to changes. Age and sex did not show a direct link to performance. However, skills and experience from longer service strongly affect how personnel perceive modernization.

Newer personnel with fewer years of service often see modernization differently. They enter the BFP with higher expectations. They adapt more easily to new technologies. De Guzman (2022) found that shorter-service personnel accept modernization programs more readily

because they are less tied to traditional practices. This generational difference creates a potential gap in acceptance and implementation. The BFP needs policies that include both older and newer personnel.

In the context of the BFP Modernization Act of 2021 (R.A. 11589), which provides a ten-year modernization plan, years of service are important for evaluation. Lopez (2022) notes that modernization should not be measured only by technology. It should also consider how personnel at different service levels respond. Including both new and seasoned staff in decision-making ensures modernization meets practical needs and plans for the future—years of service shape how personnel judge modernization programs. Senior staff rely on their experience in the BFP or volunteer unit. They compare modernization efforts to past limitations in equipment, training, and systems.

Cruz (2020) reports that long-serving personnel use outdated systems as a benchmark. They either support modernization if they see improvements or criticize it if changes seem impractical. Research shows that tenure in public service builds professional identity and culture (Miller et al., 2017). This culture shapes how personnel interpret modernization. Long-serving staff often value upgrades such as advanced fire trucks, communication systems, and emergency medical equipment. They have seen the system evolve from limited resources to modern tools (Perez et al., 2021).

Newer personnel often have different views. They adapt to technology more quickly. Delos Santos (2022) confirms that shorter-service personnel are more open to modernization because older practices have less influence on them. This difference highlights the need for inclusive policies. Years of service also affect the adaptation to training. Hernandez (2020) found that senior personnel may find digital-based modernization programs harder to learn. Training that considers different experience levels reduces resistance. It helps modernization succeed.

Finally, the BFP Modernization Act of 2021 sets a ten-year roadmap. Years of service remain a key factor in evaluating success. Lopez 2022) stresses that modernization should balance technology with staff perspectives. Including both senior and new personnel ensures practical and forward-looking results.

Fire Protection Services

Fire protection services are organized systems to prevent, respond to, and reduce fires and related emergencies. The International Fire Service Training Association (FSTA, 2020) defines fire protection as encompassing education, prevention, preparedness, and suppression. In the Philippines, the Bureau of Fire Protection (BFP) under Republic Act No. 6975 leads these services. The BFP Modernization Act of 2021 (R.A. 11589) aims to make the agency more professional and technologically advanced to handle modern fire and emergency challenges.

The law requires the establishment of fire protection services and rescue hotlines in all local government units. Municipalities without fire services receive priority (Lawphil, 2021).

Before modernization, the BFP faced many problems. The equipment was outdated. Staffing was low. Resources were scarce. The Commission on Audit (2018) reported delays in fire station projects and in the delivery of fire trucks. By mid-2018, the Bureau of Fire Protection had completed only 263 of the 945 planned stations. Ayalin (2019) added that 308 towns had no fire stations, and 59 had no fire trucks. The Philippine Star (2018) noted that the BFP lacked over 26,000 personnel and proper safety gear. Even the BFP's 2018 Accomplishment Report confirmed delays and suspended projects.

The administration unevenly distributed resources. Urban areas had more equipment and staff than rural towns. Poor facility design, weak maintenance, and inadequate communication systems lowered provincial and municipal response capabilities (DOF-BFP Report, 2019).

Firefighters struggled to reach remote areas, which increased the number of casualties. Outdated procedures and inadequate training made work more difficult. De Leon (2020) stated that many firefighters lacked access to advanced training in modern techniques, hazardous materials, and rescue operations. The absence of a centralized digital database also limited data collection and planning. Public awareness was low. Villanueva (2020) reported that fire prevention campaigns rarely reached informal settlements. Without education and local engagement, fire incidents remained frequent in dense communities.

Modernization has improved fire protection in the Philippines. Suppression and rescue operations are faster and safer. Technology such as the national 911 system, digital monitoring platforms, GIS, and drones has improved deployment and reduced response times (Delos Santos, 2022).

The BFP Modernization Act of 2021 provides legal and financial support. The BFP now acquires modern equipment, builds new fire stations, and improves nationwide communication networks. The act also focuses on recruiting, training, and developing both personnel and volunteers to improve readiness and service delivery (BFP, 2021).

The BFP protects lives and property through fire suppression, prevention, investigation, and emergency medical response (Delos Santos, 2021). These services follow the Fire Code of the Philippines (R.A. 9514), which mandates safety inspections and enforcement. Reyes (2022) added that the BFP conducts inspections, issues permits, and leads public education campaigns to prevent fires. Navarro (2020) noted that the BFP also handles rescue operations during disasters, combining firefighting with multi-hazard emergency response.

Fire prevention and investigation are key. Gonzalez (2022) reported that the Fire Safety Enforcement Division leads campaigns such as "Oplan Ligtas na Pamayanan," which reduce structural fires through household and workplace inspections. Villanueva (2021) explained that

investigations now track the causes of fires and promote accountability. Martinez (2022) emphasized that modernization provided new fire trucks, rescue vehicles, and protective gear. Tan (2022) observed improvements in emergency communication and coordination, which speed up responses during major fires and disasters.

Fire protection is crucial for community safety. The World Fire Statistics Centre (2021) shows that strong fire services reduce casualties and property loss. Modern fire services protect lives and support economic stability. DILG (2023) noted that stronger fire services help the Philippines meet Disaster Risk Reduction and Management goals. Effective fire services also build public trust and encourage participation in safety programs. Fire protection now emphasizes prevention. Programs include education, school campaigns, and community drills. Proactive measures reduce risks before emergencies occur. These efforts teach citizens safe practices and strengthen community responsibility (Haddow et al., 2021).

Cote and Bugbee (2019) highlighted that modern fire departments improve public safety through infrastructure upgrades and training. The BFP modernization program focuses on technology and efficiency to meet global standards. Haddow et al. (2017) added that fire protection supports disaster risk reduction through public awareness, drills, and hazard mitigation. Stull (2015) emphasized that well-trained and equipped fire departments reduce life and property losses while maintaining economic stability. The BFP modernization aligns with this view by upgrading equipment communication systems and protective gear.

Globally, modernization improves performance and disaster resilience. Singapore uses smart fire stations, digital dispatch, and rapid response units to shorten response times (Tanaka, 2021). Japan applies GIS and predictive fire analysis to improve deployment (Kobayashi, 2020). The United States uses standardized command systems, real-time data, and advanced training to reduce casualties (US Fire Administration, 2020). The United Kingdom uses digital fire modeling

and automated resource allocation for risk forecasting (Brown, 2021). Australia integrates climate-based fire forecasting to manage wildfires (Peters, 2022).

International evidence shows that continuous modernization improves community resilience and reduces economic losses. The US Fire Administration (2025) highlights advanced training, technology, and public fire education as essential to reducing casualties in both structural and wildfire incidents.

Force Restructuring and Organizational Development

Force restructuring and organizational development are changes to the BFP's internal structure and management. These reforms improve coordination, accountability, and efficiency. They include reorganizing departments, creating new positions, decentralizing authority, and forming specialized divisions. Burke (2020) said organizational development uses planned actions to increase an agency's capacity through continuous improvement.

Restructuring and development help public safety institutions, such as the BFP, work more effectively. Burke (2022) explained that these efforts improve effectiveness through redesign, leadership growth, and personnel training. Fire service leaders restructure their teams to distribute staff effectively and align hierarchies with operational needs.

The BFP Modernization Program aims to build a compact, efficient, and modern firefighting force under R.A. 9514. Within 90 days of the rules, the BFP must submit a revised organizational structure and staffing plan to the DILG and DBM. The BFP also establishes a qualification program for officers and personnel in accordance with Civil Service Commission rules (Lawphil, 2021).

Before modernization, the BFP faced many challenges Lantaco (2019) reported an outdated structure that limited local decision-making. Garcia (2020) noted that urban areas had

enough staff while rural stations lacked personnel and specialized units. Mendoza (2018) said unclear promotion paths lowered morale. Villanueva (2019) observed weak coordination between the administration and operations, resulting in inefficiencies. Lopez (2020) highlighted poor leadership training and a lack of ongoing development, both of which reduce adaptability.

After modernization, the BFP improved. Rivera (2022) said that restructuring regional and command offices enabled better authority distribution and faster decision-making. New divisions, such as Strategic Management and Fire and Emergency Medical Services, integrated administrative, operational, and technical tasks. Indama (2023) noted clearer hierarchies and performance-based leadership. These reforms addressed previous inefficiencies and made the BFP more adaptable.

The national government also supported capacity building. Soriano (2023) stated that DILG and CSC launched leadership and competency programs for officers. Lim (2022) added that partnerships with universities allow personnel to pursue education in public safety management and leadership. Restructuring improves efficiency and adaptability. Burke (2017) explained that redesigning structures and developing leaders and personnel enhances effectiveness. A clear structure speeds decisions, reduces redundancy, and improves accountability (North, 2022). A modern BFP can meet growing fire safety demands, especially in urban areas.

R.A. 11589 reorganizes the BFP into a responsive and professional agency. It adds positions, adjusts ranks, and creates specialized units for modern firefighting and emergency response (Congress of the Philippines, 2021). Restructuring improves operational capability and efficiency.

Heladero et.al (2025) studied BFP personnel in Southern Bukidnon. They found high effectiveness in fire suppression. They recommend adding more staff, providing ongoing training,

and defining clear roles. They also suggested competency profiling and career laddering to support a professional, credentialed force.

Other countries provide lessons. Anderson (2021) showed that Australia's Fire and Rescue Services used an Integrated Command and Control System this improved coordination and resource allocation. Yamamoto (2022) reported that Japan's local fire departments adopt collaborative models for disaster management.

The International Association of Fire Chiefs (2019) stated that restructuring improves coordination, accountability, and decision-making. Kim et al. (2020) added that it increases morale and professionalism when paired with leadership development and training. For BFP personnel and volunteers, reforms improve role clarity, supervision, and career growth.

Modernization in Europe and North America offers benchmarks. Williams (2022) noted that the UK's NFCC introduced performance-based management, which inspired the Philippines to adopt it. Johnson (2023) reported that the USFA strengthened leadership and established task forces to address emerging hazards. These examples show that ongoing modernization and learning are key to effective fire services.

Capability, Material, and Technology Development

The BFP Modernization Program, under R.A. 11589, aims to make the agency capable, efficient, and modern. It plans to upgrade equipment, vehicles, and protective gear. Old outdated equipment will be phased out (Lawphil, 2021).

Capability, material, and technology development (CMTD) are key parts of modernization. Grant (2019) says capability development improves skills, knowledge, and competence in high-risk work. Cote and Bugbee (2019) note that updating fire trucks, protective equipment, and rescue tools improves safety and response.

Technology also strengthens coordination and situational awareness. Haddow et al. (2017) found that GIS, digital command systems, and real-time communication improve emergency management worldwide. Together, capability, materials, and technology form the core of modern fire services.

The Commission on Audit (2017) reported that many BFP units used old or unserviceable fire trucks, some dating back to the 1950s. At least 349 municipalities had no fire trucks. PPE such as helmets, coats, gloves, boots, and breathing gear was also insufficient. The BFP Logistical Report (2021) showed that 197 municipalities still lacked fire stations and trucks. BARMM had 63 towns without any fire trucks. These show uneven resource distribution. The Commission on Audit reported that the Bureau of Fire Protection procured only 621 of the 1,057 planned fire trucks and completed 263 of 945 targeted stations by mid-2018. Maintenance budgets were too low to keep equipment working (Lopez, 2020). Supply-chain issues and lack of technical expertise slowed the adoption of new tools (Castillo, 2020).

Training gaps also limited the capability. Santos (2020) and Delos Reyes (2019) found personnel lacked structured programs for advanced rescue, chemical fire management, and digital systems. Many offices still relied on manual systems. R.A. 11589 directs funds to upgrade vehicles, tools, and communication while phasing out old equipment (Congress of the Philippines, 2021; BFP, 2021). Castillo (2022) said ICT systems now support dispatch and monitoring.

The DILG (2023) confirmed that it is building new fire stations, installing command systems and real-time tracking tools across the country, and working with private and international partners to fund advanced equipment and innovation. The BFP Training Institute leads capacity building. Training focuses on digital literacy, equipment handling, and international standards (Echavaria, 2022). DOST research supports locally made firefighting tools. The modernization program also mandates the use of advanced equipment. These include aerial

ladders, rescue vehicles, communication systems, and breathing apparatuses that meet international standards (BFP, 2021).

Villanueva (2022) noted that structured, rank-based training strengthens skills in fire suppression, emergency response, and disaster management. Lorenzo (2022) observed the use of drones and GIS mapping for better situational awareness. DOST (2022) helps develop local firefighting tools and detection systems. Regional training centers expand access to programs (De Guzman, 2021). International partners like JICA and NFPA provide technical support. DILG (2022) introduced an Integrated Information System that connects stations for real-time reporting and resource monitoring. These improve coordination and reduce delays.

Globally, integration capabilities and technologies improve emergency response. USFA (2019) uses simulation and leadership training to improve decision-making. The UK Fire and Rescue Service (Home Office, 2020) standardizes firefighter skills. IFSTA (2020) promotes the use of virtual reality and online simulations for training. Tanaka (2021) highlighted robotic systems and AI in Tokyo. Kim (2021) and Kobayashi (2020) noted the use of drones, IoT, and thermal imaging in Japan and South Korea. Australia and Canada use simulation centers and data analytics (Harris, 2020). FEMA (2021) in the US funds modern tools like GIS, sensors, and drones. These examples show that continuous innovation and investment in technology are essential for readiness and public safety.

Specialized Services Development

Specialized Services Development focuses on improving advanced teams that handle high-risk emergencies. These teams need special training, equipment, and coordination (Harison, 2018). Under the BFP Modernization Program, specialized teams include HAZMAT, Urban Search and Rescue (USAR), Emergency Medical Services (EMS), and Fire and Arson Investigation Units

(BFP, 2022). The 2021 BFP Modernization Act also covers units for high-rise fires, forest fires, aircraft fires, chemical incidents, and medical emergencies (Lawphil, 2021).

Castro (2022) said these units strengthen BFP's ability to respond to industrial explosions, building collapses, and large disasters. Santos and Mendoza (2023) noted that EMS divisions with modern ambulances and trained paramedics now provide pre-hospital medical care as part of fire operations. Ogle et.al (2021) highlighted the creation of the Fire Arson Investigation Division (FAID), improving investigations through science and agency coordination. These changes show that BFP is now a modern, multi-skilled emergency agency.

Before 2021, BFP struggled with specialized services. Funding, training, and equipment were limited. Baluran (2018) found that BFP lacked expertise and resources for HAZMAT, USAR, and EMS, resulting in slow, inefficient responses. Navarro (2019) added that outdated rescue tools forced firefighters to resort to manual methods, thereby increasing risks.

The Commission on Audit (2018) found incomplete equipment purchases, delayed projects, and gear shortages. Many stations could not form complete rescue or HAZMAT teams. García (2019) noted that unclear guidelines caused coordination problems with other agencies. Lopez (2020) found that there were no standardized protocols or certifications, leading to inconsistent performance across regions.

Manalo (2018) stated that BFP had limited collaboration with international fire organizations, thereby restricting the adoption of best practices. Villanueva (2020) added that bureaucracy and budget limits slowed equipment purchases. Reyes and Alonzo (2020) said that the lack of training facilities and instructors prevented skill development. Morales (2021) warned that these gaps increase risks for responders and the public. These studies show that modernization was urgently needed.

Implementation of Specialized Services Development follows a phased, strategic plan. Perez (2022) stated that specialized units were deployed to high-risk areas to respond quickly. Garcia (2023) reported that new regional training centers now provide ongoing learning and certification.

Technology plays a key role. Nakamura (2022) said thermal cameras, GIS fire maps, and digital communication improve operations. Alvarez (2023) noted that partnerships with local governments and private groups improved logistics, funding, and coordination.

Lopez and Santiago (2023) emphasized the need for support programs for firefighters physical and mental health. These measures show that modernization has successfully made Specialized Services Development operational. BFP is now a professional, efficient, multi-skilled emergency agency.

Globally, specialized services are central to modern fire management. Reuters (2025) reported that Greece strengthened specialized firefighting units to respond faster to wildfires. UNDP has helped countries build specialized fire units with modern tools, training, and planning, similar to what the Philippines has done.

Thompson (2022) said the Philippines' strategy matches global best practices seen in Singapore and Tokyo. Luga et.al (2023) added that adopting Incident Management Systems (IS) and standard protocols brings BFP closer to IFSAC standards.

Nationally, R.A. 11589 created USAR, HAZMAT, EMS, and Fire Arson Investigation units (BFP, 2022). Rodriguez (2022) said the law provided funding and legal support to equip these units. Santos and Velasco (2023) noted that JICA and USAID partnerships improved skills through specialized training and field exercises. Martinez (2022) highlighted mobile command centers, rescue boats, and drones that improve situational awareness.

Villanueva (2023) and Lopez (2022) reported specialized programs for high-angle, confined-space, and water rescues. The Office of Civil Defense (OCD, 2022) reported improved coordination with other agencies. The Bureau of Fire Protection (BFP) adopted NFPA (2020) guidelines to professionalize specialized responders. Villamor and Cruz (2023) explained that sustained funding supports continuous advanced training for firefighters.

Human Resources Development

The BFP Modernization Program focuses on developing a complete training system for both personnel and volunteers. These include mandatory training for new staff and advanced training facilities (Lawphil, 2021). Human Resource Development (HRD) is central to this plan. It aims to improve skills, professionalism, and efficiency among public safety personnel. Swanson and Holton (2019) explained that HRD develops human potential through training, career growth, and organizational improvement to meet individual and institutional goals. In government modernization, Werner and DeSimone (2012) noted that continuous HR training helps employees adapt to new technology, management systems, and policies.

The BFP Modernization Act of 2021 (R.A. 11589) emphasizes HRD as key to strengthening operational capacity. It established the BFP Training Institute, professional education programs, and leadership development initiatives. Before 2021, BFP faced many HRD challenges. Fernandez (2018) noted that training was inconsistent across regions. Domingo and Perez (2019) reported outdated materials and a lack of modern facilities, which limited career growth and emergency preparedness.

Aguilar (2020) found that limited funding and few training opportunities led to delayed promotions, low motivation, and poor retention. Castillo (2019) highlighted unclear career

pathways, which weaken leadership development. Reynoso (2020) said poor coordination with other agencies limited inter-agency training.

External pressures added more problems. Cruz (2020) noted that frequent fire disasters and new hazards required skills that many personnel lacked. Espinoza (2021) said that missing stress management training led to burnout and low morale. Villareal (2020) noted that gender inclusivity in training was often overlooked, thereby limiting opportunities for women.

The implementation of HRD under R.A. 11589 transformed BFP. The BFP Training Institute now offers structured programs covering technical, managerial, and leadership skills (BFP, 2021). Digital learning and virtual simulations improve access and modernize instruction (Cruz, 2023).

Collaboration is important. Reyes and Bautista (2023) documented partnerships with the CSC, DI G, and universities for specialized leadership courses. The DOST (2022) co-developed fire safety technologies to integrate science into training.

Fernandez (2021) noted that HRD now includes welfare programs like mental health counseling, insurance, and family support. Villanueva (2023) said sustainability requires adapting global best practices to local needs. Garcia (2023) added that digital performance monitoring ensures transparency, consistency, and accountability.

Globally, fire services see HRD as an investment. In Australia, the CFA has yearly competency checks, leadership workshops, and mental health programs (Harris, 2020). Canada's Fire Service College includes diversity, wellness, and psychological readiness. Japan's FDMA uses tiered technical, leadership, and psychological training for lifelong learning (Kitagawa, 2020). In the US, HRD is standardized. Hulett (2018) explained that the USFA and NFA offer leadership, identification management, and technical rescue training. FEMA (2019) supports certification systems like the Pro Board to ensure consistent competency across fire departments. These

practices show that structured HRD improves professionalism, reduces training gaps, and enhances service goals, as mirrored by BFP modernization.

Locally, the Philippines has adopted HRD reforms based on international standards. Dela Peña (2022) said the BFP Training Institute now offers competency-based programs in Fire Investigation, EMS, and USAR. Bradley (2021) noted that leadership immersion and merit-based promotion improved morale and career satisfaction.

Garcia and Mendoza (2023) said HRD also promotes continuous education and technical certifications, such as HAZMAT and Fire Safety Inspection Training. Lopez (2022) added that these programs build a skilled workforce ready for complex fire and rescue tasks. The CSC (2022) recognized the importance of partnerships with universities and agencies to strengthen leadership, accountability, and ethics.

Finally, Cruz (2023) and Villareal and Santos (2023) reported that digital learning and simulation-based training now align BFP HRD with global fire education standards. Bautista (2022) added that programs emphasize values and professionalism, ensuring personnel uphold service excellence and ethics.

Theoretical Framework

This study draws on key ideas from organizational change, technology adoption, and employee motivation. These theories help explain how the 2021 modernization programs may be affecting the Bureau of Fire Protection (BFP) and the volunteer firefighters who support it. By looking at these perspectives, we can better understand how modernization influences not only the tools and processes but also the people who make the fire service work.

This study is anchored on the Modernization Theory developed by Walt Whitman Rostow (1960). Rostow's theory explains how organizations and societies progress through stages of development from being traditional and less efficient to becoming modern, dynamic, and highly capable. He believed that modernization occurs when institutions adopt new technologies, improve their systems, and enhance their personnel, resulting in greater productivity and progress.

In relation to this study, Rostow's Modernization Theory helps explain how the 2021 Bureau of Fire Protection (BFP) Modernization Program aims to transform the agency into a more advanced and efficient organization. The program's components, such as upgrading fire equipment, improving facilities, providing continuous training, and developing specialized services, reflect the process of modernization that Rostow described. These efforts signify the BFP's movement toward a more professional and well-equipped institution capable of delivering better public service.

The relevance of this theory lies in showing how modernization can influence the performance and perception of BFP personnel. The independent variable, which is the 2021 Modernization Program, includes improvements in fire protection services, force restructuring and organizational development, capability, material and technology development, specialized services development, and human resource development. The dependent variable, which is the

perception of BFP personnel, reflects how these modernization efforts are experienced by those working within the organization.

Applying Rostow's Modernization Theory suggests that as the BFP continues to modernize through better tools, training, and systems, its personnel are likely to feel more motivated, capable, and confident in performing their duties. In this way, modernization becomes not just an institutional goal but a direct factor in improving employee morale and service quality.

Overall, Rostow's Modernization Theory serves as the guiding framework for this study. It supports the idea that modernization programs, like that of the BFP, play a vital role in driving institutional progress and shaping how personnel perceive and experience change within their organization.

Schematic Diagram

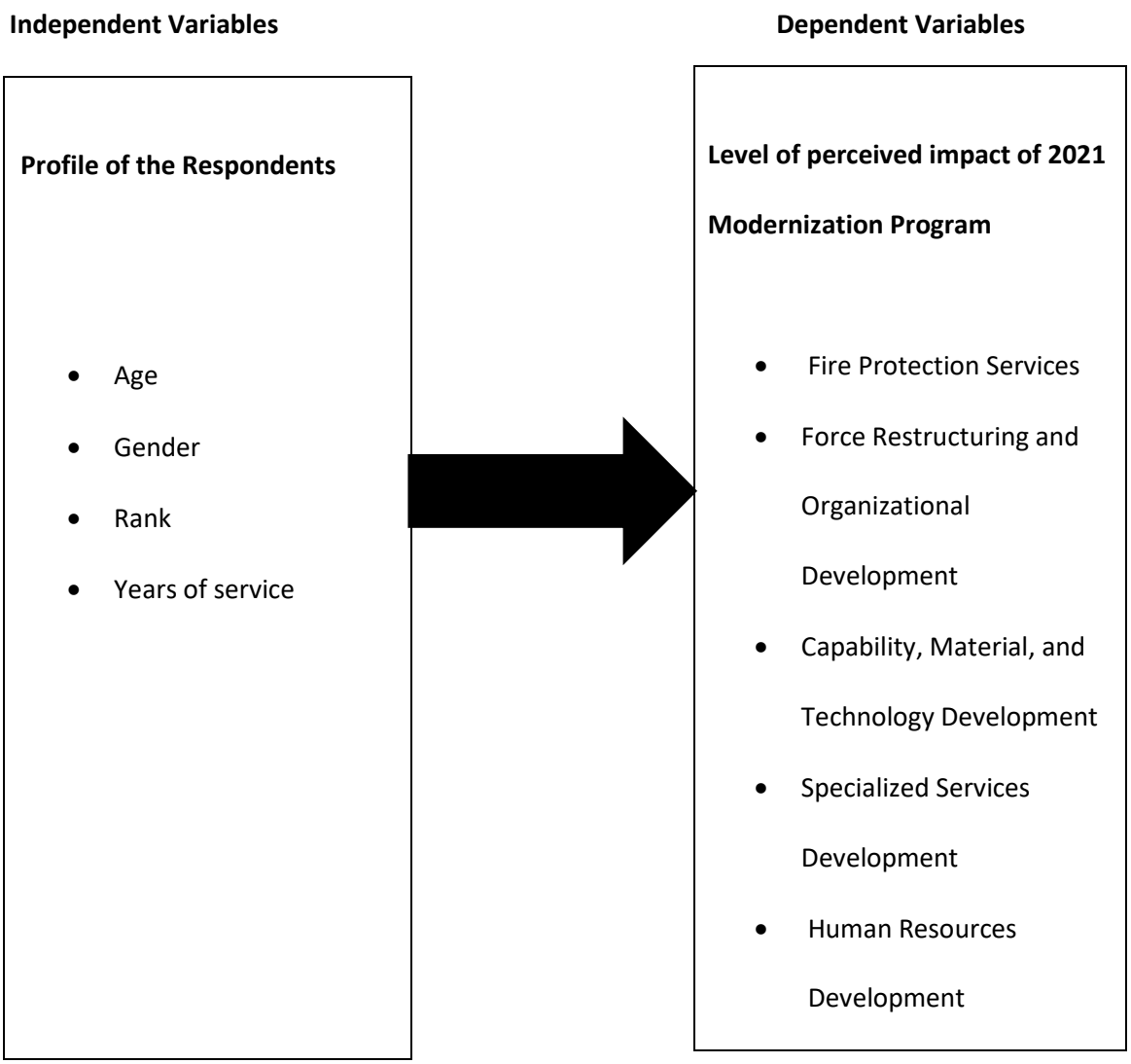


Figure 1. The Schematic Diagram shows the relationship between the dependent and independent variables of the study.

Statement of the Problem

This study aims to determine the perceived impact of the 2021 Modernization Program among BFP personnel and fire volunteers, considering their demographic profiles. Specifically, it seeks to answer the following questions:

1. What is the profile of the respondents in terms of:

   1.1 Age

   1.2 Gender

   1.3 Rank

   1.4 Years of service

2. What is the level of perceived impact of the respondents on the 2021 Modernization Program in terms of:

   2.1 Fire Protection Services

   2.2 Force Restructuring and Organizational Development

   2.3 Capability, Material, and Technology Development

   2.4 Specialized Services Development

   2.5 Human Resources Development

3. Is there a significant relationship on the perceived impact of the 2021 Modernization Program when grouped according to their profile?

Hypothesis

H0: There is no significant relationship in the perceived impact of the 2021 Modernization Program when grouped according to profile.

Definition of Terms

The following terms are defined both operationally and conceptually for the purposes of this study:

Age. It refers to how long a particular respondent has lived.

Gender. Refers to identifying a person as male, female, or other, thereby helping recognize gender differences.

Rank. Refers to the person who holds a position or job within the Bureau of Fire Protection, or as a volunteer, such as a firefighter, officer, or team leader, which might affect how they experience the modernization.

Years of Service. Refers to how long a person has worked or volunteered with the firm, giving insight into whether more experienced members view changes differently from newer ones.

Fire Protection Services. Refers to enhancing fire prevention, suppression, and emergency response capabilities through better coverage and the establishment of rescue hotlines in all local government units in Buenavista, Nasipit, and Carmen.

Force Restructuring and Organizational Development. Refers to the reorganization of the BFP's structure and workforce to improve operational efficiency and align with modernization goals through optimized command, roles, and internal processes.

Capability, Material, and Technology Development. Refers to the acquisition and deployment of advanced firefighting equipment, technology, and materials to enhance the operational effectiveness and technological edge of the BFP.

Specialized Services Development. Refers to the enhancement of specialized firefighting and emergency services, including chemical, biological, radiological, nuclear, and explosive emergency response capabilities, supported by expert services and advanced coordination systems.

Human Resources Development. Refers to the continuous training, professional development, and capacity building of BFP personnel and volunteers to equip them with advanced skills and knowledge necessary for effective fire and emergency operations.

CHAPTER 2

RESEARCH METHODOLOGY

This chapter covers the methods used for data collection and analysis in the present study. It explains the various stages of the research, which include the research design, research materials and methods, research environment, population and sample, research data and analysis plan, and research ethical standards.

Research Design

This study employed a descriptive–correlational quantitative research design. The descriptive part focused on identifying the demographic profile and perceptions of Bureau of Fire Protection (BFP) personnel regarding the 2021 Modernization Program. The correlational part aimed to determine whether there is a significant relationship between the perceived implementation of the modernization program and the perceived impact on fire protection services. A correlational design is appropriate when the researcher seeks to determine the degree of relationship between two or more variables without manipulating them. This design allows the study to identify the association between modernization and its perceived effects on BFP operations while maintaining objectivity using numerical data (Creswell, 2018).

To represent the respondents, the researchers used purposive sampling. This sampling approach ensures that the researchers intentionally select participants based on their specific characteristics or experience that are relevant to the research question. By using this method, the researchers obtained more accurate and insightful information, enabling them to draw stronger and more meaningful conclusions about the impact of modernization on fire protection services.

Research Materials and Methods

This study used a quantitative approach. A survey questionnaire served as the main instrument for data collection. The researchers divided the questionnaire into two main sections. The first section collected information on respondents' demographic profiles, including age, gender, rank or volunteer role, and years of service.

The second section focused on their perceptions of the BFP Modernization Program, covering five key areas: Fire Protection Services; Force Restructuring and Organizational Development; Capacity, Material, and Technology Development; Specialized Services Development; and Human Resource Development. The researchers applied a four-point Likert scale, with four indicating Strongly Agree, three indicating Agree, two indicating Disagree, and one indicating Strongly Disagree. This approach helped the researchers better understand how BFP personnel and fire volunteers view the organization's modernization efforts.

Research Environment, Population, and Sample

Research Environment

The researchers conducted the study in Buenavista, Nasipit, Carmen, and Butuan City in Agusan del Norte, Philippines. The study was conducted specifically in Buenavista, Nasipit, Carmen, and Butuan City. The location is densely populated, featuring a mix of residential, commercial, and industrial areas. It has a diverse population, including BFP personnel. The researchers chose Buenavista, Nasipit, Carmen, and Butuan City to collect responses, allowing them to capture diverse perspectives on the BFP modernization program, making these locations ideal for the study.

Population and Sample

Table 1. Respondents of the study

Respondents	Population	
	F	%
BFP NASIPIT	15	13.2
BFP CARMEN	13	11.4
BFP BUENAVISTA	20	17.5
BFP LIBERTAD	11	9.7
BFP BUTUAN CITY	47	41.2
BFP AMPAYON	8	7.0
Total	**114**	**100.0**

Table 1 shows the population of the BFP NASIPIT, BFP CARMEN, BFP BUENAVISTA, BFP LIBERTAD, BFP BUTUAN CITY, AND BFP AMPAYON.

Research Data and Analysis Plan

The researchers first secured approval from the research adviser and the dean of the CCJE Department. After the approval, we sent a Letter of Request to the Municipal Fire Marshals of Buenavista, Nasipit, and Carmen, seeking permission to obtain the total number of BFP personnel and fire volunteers for the survey. The Municipal Fire Marshals approved the request, allowing the researchers to proceed with the study. Next, we distributed the questionnaires face-to-face to the selected respondents in the three municipalities. After collecting, the research team reviewed each questionnaire for completeness and accuracy, returning any incomplete forms for clarification where necessary. Finally, we will consolidate and process the data. Responses will be checked, tallied, tabulated, and analyzed using frequency and percentage, as outlined in SOP1. In SOP2, we will use the mean, and in SOP3, we will use the pearson correlation coefficient.

Research Ethical Standards

A. Protection of Intellectual Rights

The researchers ensured they respected others' intellectual property by properly citing all sources used in the study. Ideas, concepts, and materials from other authors were acknowledged to uphold honesty and fairness in academic work.

B. Informed Consent

Before participating in the study, the researchers fully informed the respondents about its purpose, goals, and procedures. The researcher explained the participants' roles, described how they would collect and store the data, and clarified how they would use the information. The study included only participants who gave informed consent.

C. Data Privacy and Confidentiality

Protecting respondents' privacy was a top priority. The researchers kept the information from the questionnaires strictly confidential and used it only for research purposes.

D. Voluntary Participation and Freedom to Withdraw

Participation in the study was entirely voluntary. Respondents could refuse to take part or withdraw at any time without facing any penalties.

E. Minimization of Harm and Risk Management

The researchers ensured that no participant experienced any form of physical, emotional, or psychological harm during the study. The researchers made every effort to maintain their comfort, safety, and well-being throughout the process.

F. Beneficence and Contribution to Knowledge

This research aimed to generate valuable insights to improve fire protection services and modernization programs. The study also hoped to contribute to the broader field of public safety and community development.

G. Justice and Fair Participant Selection

Participants were chosen fairly and based on their relevance to the study. No one was excluded or discriminated against. Each respondent received an equal opportunity to participate in the research.

H. Data Integrity and Accuracy

The researchers handled all gathered data carefully and reported the results truthfully. The researchers ensured that the results presented were accurate, reliable, and free of manipulation.

I. Transparency and Honesty in Reporting

The researchers were transparent about all procedures, results, and limitations of the study. The researchers openly acknowledged any possible bias or error to maintain the study's credibility.

J. Use of Patented or Copyright Materials

All copyrighted or patented materials, such as books, questionnaires, and digital tools, were cited correctly and used only with permission. The researchers always respected intellectual property rights.

K. Ethical Considerations for Animal and Human Traits

Since the study involved human participants, their rights, dignity, and welfare were always respected. The researchers tolerated no form of exploitation or harm throughout the research.

L. Responsible Use of AI and Other Related Technologies

The researchers used artificial intelligence and digital tools to process data or prepare content. They applied these technologies responsibly and ethically to preserve the study's originality and integrity.

M. Ethical Clearance and Institutional Approval

Before the study began, the researchers secured the necessary approvals and clearances from the appropriate authorities. These ensured that all research activities followed institutional and academic ethical standards

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ORGANIZATION	POSITIONS HELD

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Membership/Affiliations

ORGANIZATION	POSITIONS HELD